Regulation of Metabolism

Human Physiology - Chapter 19

Teaching Resource

December 24, 2025

Outline

19.1 Nutritional Requirements

19.2 Regulation of Energy Metabolism

19.3 Energy Regulation by the Pancreatic Islets

19.4 Diabetes Mellitus and Hypoglycemia

19.5 Metabolic Regulation by Other Hormones

19.6 Regulation of Calcium and Phosphate Balance

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Metabolic Rate and Energy Balance

- **Metabolic rate** (代谢率): The rate of energy expenditure measured by oxygen consumption
- **Basal metabolic rate (BMR, 基础代谢率)**: Energy expenditure at rest, 12-14 hours after eating
  - Influenced by age, sex, body surface area, and thyroid hormone levels
- Energy balance:
  - **Positive balance** (正能量平衡): Intake > expenditure → weight gain
  - **Negative balance** (负能量平衡): Intake < expenditure → weight loss
  - Body defends usual weight by adjusting metabolic rate

Metabolic Rate and Energy Balance

- Energy expenditure varies greatly with physical activity level
- Understanding caloric costs helps in designing exercise programs for weight management
- Activities range from low-intensity (2-3 kcal/min) to high-intensity (12-20 kcal/min)

Anabolic Requirements

- **Anabolism** (合成代谢): Synthesis of complex molecules (proteins, glycogen, triglycerides)
- **Catabolism** (分解代谢): Breakdown of molecules for energy
- **Turnover rate** (周转率): Rate of breakdown and resynthesis
- **Essential amino acids** (必需氨基酸): Nine amino acids that must be obtained from diet
- **Essential fatty acids** (必需脂肪酸): Linoleic and linolenic acids
  - **Omega-3 fatty acids** (ω-3 脂肪酸): Protective against cardiovascular disease

Vitamins: Classification and Functions

- **Vitamins** (维生素): Small organic molecules serving as coenzymes
- Two classes:
  - **Fat-soluble** (脂溶性): A, D, E, K
  - **Water-soluble** (水溶性): B-complex and vitamin C
- Requirements vary by age, gender, and physiological state

Vitamins: Classification and Functions

- Recommended Dietary Allowances (RDA, 推荐膳食摄入量) provide guidelines for adequate nutrition
- Values differ for children, adults, pregnant women, and lactating women
- Both deficiency and excess can have adverse health effects

Vitamins: Classification and Functions

- Each vitamin has specific dietary sources and metabolic functions
- Deficiency symptoms help identify nutritional inadequacies
- Fat-soluble vitamins (A, D, E, K) can be stored and may accumulate to toxic levels
- Water-soluble vitamins (B-complex, C) are not stored and require regular intake

Free Radicals and Oxidative Stress

- **Free radicals** (自由基): Molecules with unpaired electrons
- **Reactive oxygen species (ROS, 活性氧)**: Oxygen-containing free radicals
    - Superoxide (O_2^{.-}), hydroxyl radical (OH^{.}), hydrogen peroxide (H_2O_2)
- Normal cellular respiration produces ROS as byproducts
- **Oxidative stress** (氧化应激): Excessive ROS damages DNA, proteins, and lipids

Free Radicals and Oxidative Stress

- Balance between ROS production and antioxidant defenses is crucial
- Enzymatic defenses: **SOD** (超氧化物歧化酶), **catalase** (过氧化氢酶), **glutathione peroxidase** (谷胱甘肽过氧化物酶)
- Nonenzymatic defenses: **Glutathione** (谷胱甘肽), **vitamin C** (抗坏血酸), **vitamin E** (生育酚)
- Excessive ROS causes cellular damage; inadequate ROS impairs immune function

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Circulating Energy Substrates

- Blood carries energy substrates to tissues for cell respiration
- Different organs have different preferred energy sources:
  - **Brain**: Requires glucose (except during prolonged fasting when it can use ketones)
  - **Skeletal muscle**: Can use glucose, fatty acids, or ketone bodies
  - **Liver**: Oxidizes fatty acids and amino acids

Circulating Energy Substrates

- Energy reserves (glycogen, triglycerides, proteins) are hydrolyzed to produce circulating substrates
- Circulating substrates (glucose, fatty acids, amino acids) are available for immediate use
- All substrates ultimately enter aerobic respiration pathways to generate ATP
- This diagram shows the interconversion and flow of energy-storage molecules

Adipose Tissue as an Endocrine Organ

- **Adipose tissue** (脂肪组织): Not just passive storage, but active endocrine organ
- **Adipokines** (脂肪因子): Hormones secreted by fat cells
  - **Leptin** (瘦素): Signals energy stores, reduces appetite, increases metabolic rate
  - **Adiponectin** (脂联素): Enhances insulin sensitivity, decreases with obesity
  - **TNF-α**: Pro-inflammatory, increases with obesity, contributes to insulin resistance
- **Brown adipose tissue (BAT, 棕色脂肪组织)**: Produces heat through **uncoupling protein 1 (UCP1)**

Adipose Tissue as an Endocrine Organ

- **Leptin** (瘦素) is secreted by adipocytes in proportion to fat mass
- Crosses blood-brain barrier to affect **arcuate nucleus** (弓状核) neurons in hypothalamus
- Reduces appetite and increases metabolic rate, promoting energy balance
- Insulin stimulates leptin secretion from adipose tissue
- **Leptin resistance** in obesity may impair weight regulation

Hormonal Signals Regulating Appetite

- Multiple hormones coordinate to regulate feeding behavior
- Short-term signals from gastrointestinal tract:
  - **Ghrelin** (胃饥饿素): Stimulates hunger before meals
  - **CCK** (胆囊收缩素): Promotes satiety after meals
  - **PYY**: Inhibits appetite
  - **GLP-1**: Promotes satiety and insulin secretion
- Long-term signals: Leptin and insulin signal energy stores

Hormonal Signals Regulating Appetite

- The CNS integrates multiple hormonal signals to regulate energy balance
- Inhibitory signals (red): CCK, PYY, leptin, insulin reduce appetite
- Stimulatory signal (green): Ghrelin increases appetite before meals
- Pancreatic insulin provides information about fed state
- This complex system maintains energy homeostasis

Components of Total Energy Expenditure

- Total daily energy expenditure (TDEE) has three components:
  1. **Basal metabolic rate** (基础代谢率, BMR): 60-75% of total
  2. **Physical activity**: 15-30% of total
  3. **Adaptive thermogenesis** (适应性产热): 10-15% of total

Components of Total Energy Expenditure

- Energy intake through eating must balance energy expenditure to maintain weight
- BMR is the largest component of energy expenditure at rest
- Exercise increases energy expenditure proportionally
- **Adaptive thermogenesis** includes diet-induced thermogenesis and cold-induced thermogenesis
- **Brown adipose tissue** (BAT) produces heat when activated by cold or diet

Metabolic Balance: Anabolism vs Catabolism

- Metabolic state is determined by the balance of hormones
- **Anabolic hormones** (合成代谢激素): Promote synthesis and storage
  - Insulin is the primary anabolic hormone
- **Catabolic hormones** (分解代谢激素): Promote breakdown and mobilization
  - Glucagon, glucocorticoids, epinephrine
- **Growth hormone** and **thyroxine** have both anabolic and catabolic effects

Metabolic Balance: Anabolism vs Catabolism

- The metabolic balance can be tilted toward anabolism or catabolism
- **Insulin** promotes energy storage: glycogenesis, lipogenesis, protein synthesis
- **Glucagon**, **glucocorticoids**, **epinephrine** promote energy mobilization
- **Growth hormone** and **thyroxine** have dual effects depending on tissue and conditions
- This diagram shows which hormones favor anabolic vs. catabolic pathways

Hormonal Regulation of Metabolism

- Multiple hormones coordinate metabolic responses
- Hormones can work:
  - **Synergistically** (协同作用): Amplify each other's effects
  - **Antagonistically** (拮抗作用): Oppose each other's effects
- This coordination maintains glucose homeostasis and energy balance

Hormonal Regulation of Metabolism

- Different hormones affect carbohydrate, protein, and lipid metabolism
- **Insulin**: Lowers blood glucose, promotes storage
- **Glucagon**: Raises blood glucose, promotes mobilization
- **Growth hormone**, **glucocorticoids**, **epinephrine**: Generally raise blood glucose
- **Thyroid hormones**: Increase metabolic rate, affect all pathways

Hormonal Regulation of Metabolism

- Hormones interact through complex networks with stimulatory (⊕) and inhibitory (⊖) effects
- Some processes (like glycogenesis) are promoted by insulin and inhibited by glucagon
- Other processes (like lipolysis) are inhibited by insulin and promoted by multiple hormones
- Understanding these interactions is essential for treating metabolic diseases

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Pancreatic Islets Structure

- **Pancreatic islets (Islets of Langerhans, 胰岛)**: Endocrine tissue in pancreas
  - **Beta cells** (β 细胞): Secrete insulin (70% of cells)
  - **Alpha cells** (α 细胞): Secrete glucagon (20% of cells)
  - Delta cells: Secrete somatostatin
  - PP cells: Secrete pancreatic polypeptide
- Insulin and glucagon are the primary regulators of glucose homeostasis

Mechanism of Insulin Secretion

- Insulin secretion is tightly regulated by blood glucose levels
- Glucose enters beta cells through **GLUT2** transporters
- Increased glucose metabolism raises ATP production
- Changes in cellular metabolism trigger insulin release

Mechanism of Insulin Secretion

Glucose-stimulated insulin secretion involves a sequence of events:

1. Elevated blood glucose increases glucose entry and ATP production in beta cells
2. Increased ATP closes K^+ channels, causing K^+ accumulation
3. Membrane depolarization (去极化) opens voltage-gated $\mathrm{Ca\hat{}\{2+\}}$ channels

Mechanism of Insulin Secretion

4. $\mathrm{Ca\hat{}\{2+\}}$ influx triggers exocytosis (胞吐作用) of insulin-containing vesicles

Metabolic States: Absorptive vs Postabsorptive

- **Absorptive state** (吸收状态): During and after meals (2-4 hours)
  - High insulin/glucagon ratio
  - Promotes energy storage (anabolism)
- **Postabsorptive state** (吸收后状态): Between meals and during fasting
  - Low insulin/glucagon ratio
  - Promotes energy mobilization (catabolism)

Metabolic States: Absorptive vs Postabsorptive

- During fasting, hormonal changes favor catabolism to maintain blood glucose
- **Decreased insulin** and **increased glucagon** promote breakdown of energy stores
- Liver releases glucose from glycogenolysis (糖原分解) and gluconeogenesis (糖异生)
- Adipose tissue releases fatty acids from lipolysis (脂解)
- Liver produces ketone bodies (酮体) from fatty acids as alternative fuel

Metabolic States: Absorptive vs Postabsorptive

- The inverse relationship between insulin and glucagon regulates metabolic balance
- During feeding (absorptive state): insulin ↑, glucagon ↓ → anabolism
- During fasting (postabsorptive state): insulin ↓, glucagon ↑ → catabolism
- This reciprocal regulation ensures continuous glucose supply to brain
- Maintains stable blood glucose despite intermittent food intake

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Diabetes Mellitus Overview

- **Diabetes mellitus** (糖尿病): Metabolic disease characterized by hyperglycemia
- Two major types:
  - **Type 1 (T1DM)**: Autoimmune destruction of beta cells, absolute insulin deficiency
  - **Type 2 (T2DM)**: Insulin resistance with relative insulin deficiency
- Classic symptoms: polyuria (多尿), polydipsia (多饮), polyphagia (多食), weight loss
- Chronic complications: Damage to eyes, kidneys, nerves, blood vessels

Diabetes Mellitus Overview

- Uncorrected insulin deficiency leads to a cascade of metabolic problems
- **Hyperglycemia** (高血糖) results from decreased glucose uptake and increased production
- Excessive lipolysis produces ketone bodies, leading to **ketoacidosis** (酮症酸中毒)
- **Osmotic diuresis** from glycosuria (糖尿) causes dehydration and electrolyte loss
- Without treatment, can progress to diabetic coma (糖尿病昏迷) and death

Oral Glucose Tolerance Test

- **Oral glucose tolerance test (OGTT, 口服葡萄糖耐量试验)**: Diagnostic tool
- Measures glucose and insulin responses after glucose ingestion
- Helps identify:
  - Normal glucose tolerance
  - **Prediabetes** (前糖尿病): Impaired glucose tolerance, insulin resistance
  - **Type 2 diabetes**: Frank hyperglycemia with insufficient insulin response

Oral Glucose Tolerance Test

- OGTT reveals progressive metabolic dysfunction
- **Normal**: Glucose returns to baseline within 2 hours, appropriate insulin response
- **Prediabetic**: Elevated glucose, exaggerated insulin response (compensating for insulin resistance, 胰岛素抵抗)
- **Type 2 diabetes**: Markedly elevated glucose, insufficient insulin response (beta cell failure)

Oral Glucose Tolerance Test

- Early detection allows intervention to prevent progression

Oral Glucose Tolerance Test

- Type 1 and type 2 diabetes differ in multiple aspects
- Age of onset, etiology, insulin levels, and treatment differ between types
- Type 1: Usually childhood onset, autoimmune, requires insulin therapy
- Type 2: Usually adult onset, associated with obesity, progressive disease
- Understanding differences is crucial for appropriate management

Reactive Hypoglycemia

- **Hypoglycemia** (低血糖): Blood glucose < 70 mg/dL
- **Reactive hypoglycemia** (反应性低血糖): Occurs 2-5 hours after meals
- Caused by excessive insulin secretion in response to glucose load
- More common in prediabetic individuals with insulin resistance

Reactive Hypoglycemia

- Reactive hypoglycemia shows exaggerated insulin response
- Blood glucose rises normally after glucose ingestion
- Excessive insulin secretion causes glucose to fall below normal range
- Symptoms: tremor, anxiety, sweating, weakness, hunger, confusion
- Can occur in people with insulin resistance as beta cells overcompensate

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Epinephrine and Glucagon: Shared Mechanism

- **Epinephrine** (肾上腺素) and **glucagon** (胰高血糖素) activate similar pathways
- Both promote energy mobilization during stress or fasting
- Work through **cAMP second messenger system** (cAMP 第二信使系统)
- Activate **protein kinase** (蛋白激酶) to phosphorylate target enzymes

Epinephrine and Glucagon: Shared Mechanism

- Epinephrine and glucagon share a common signaling mechanism:
  1. Hormone binds to G-protein coupled receptor
  2. G-protein dissociates, alpha subunit activates **adenylate cyclase** (腺苷酸环化酶)
  3. cAMP production activates **protein kinase A**
  4. Protein kinase phosphorylates enzymes, promoting glycogenolysis and lipolysis

Epinephrine and Glucagon: Shared Mechanism

- This mechanism amplifies the hormonal signal for rapid metabolic response

Glucocorticoids: Stress Hormones

- **Glucocorticoids** (糖皮质激素): Primarily cortisol (hydrocortisone, 皮质醇)
- Secreted by adrenal cortex in response to stress
- Promote catabolism to provide energy substrates
- Have anti-inflammatory and immunosuppressive effects
- Essential for responding to physiological stress

Glucocorticoids: Stress Hormones

- Glucocorticoids coordinate multiple catabolic processes:
- Promote protein catabolism in muscles (provides amino acids for gluconeogenesis)
- Stimulate **gluconeogenesis** (糖异生) in liver from amino acids
- Increase lipolysis in adipose tissue (provides fatty acids)
- Reduce glucose uptake by peripheral tissues (spares glucose for brain)
- Net effect: Raise blood glucose and provide alternative energy sources

Growth Hormone: Dual Effects

- **Growth hormone (GH, 生长激素)**: Secreted by anterior pituitary
- Has both direct metabolic effects and indirect growth-promoting effects
- Direct effects: Lipolysis, decreased glucose uptake (anti-insulin)
- Indirect effects: Mediated by **IGF-1** (insulin-like growth factor 1, 胰岛素样生长因子)
- IGF-1 produced by liver promotes protein synthesis and growth

Growth Hormone: Dual Effects

- Growth hormone has complex, dual metabolic effects:
- **Direct effects**: Promote lipolysis, decrease glucose uptake (catabolic, energy-mobilizing)
- **Indirect effects through IGF-1**: Promote protein synthesis, cell division, bone growth (anabolic)
- This combination is “protein-sparing”: Uses fat for energy while promoting protein synthesis
- Essential for normal growth and development in children

Acromegaly: Excess Growth Hormone

- **Acromegaly** (肢端肥大症): Results from excess GH secretion in adults
- Caused by GH-secreting pituitary tumor
- After epiphyseal plates close, GH causes enlargement of:
  - Hands and feet
  - Facial features (jaw, brow, nose)
  - Soft tissues

Acromegaly: Excess Growth Hormone

- Acromegaly demonstrates the effects of chronic excess growth hormone
- Progressive coarsening and enlargement of facial features over years
- Hands and feet also enlarge significantly
- May cause joint problems, cardiovascular disease, diabetes
- Treatment involves removing or suppressing the pituitary tumor

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Bone Remodeling

- Bone is dynamic tissue, constantly remodeling
- Two cell types with opposing functions:
  - **Osteoblasts** (成骨细胞): Build bone by depositing minerals
  - **Osteoclasts** (破骨细胞): Break down bone by resorption
- Balance between deposition and resorption determines bone mass
- Calcium and phosphate homeostasis depends on bone remodeling

Bone Remodeling

- Osteoclasts resorb bone through a two-step process:
  1. **Demineralization**: Secrete acid (H^+) into resorption pit (吸收陷窝) to dissolve $\mathrm{CaPO_4}$
  2. **Matrix digestion**: Secrete proteolytic enzymes (**cathepsin K**, 组织蛋白酶 K) to break down collagen
- This releases $\mathrm{Ca}\hat{}\{2+\}$ and $\mathrm{PO_4}\hat{}\{3\text{-}\}$ into blood

Bone Remodeling

- Process is tightly regulated by hormones to maintain calcium homeostasis

Parathyroid Hormone Regulation

- **Parathyroid hormone (PTH, 甲状旁腺激素)**: Primary regulator of calcium homeostasis
- Secreted by parathyroid glands in response to low blood $\mathrm{Ca}\hat{}\{2+\}$
- Acts on bone, kidney, and indirectly on intestine
- Works in concert with vitamin D to maintain calcium balance

Parathyroid Hormone Regulation

- PTH secretion is controlled by negative feedback:
- Low blood $\mathrm{Ca}\hat{}\{2+\}$ stimulates PTH secretion from parathyroid glands
- PTH increases bone resorption, kidney $\mathrm{Ca}\hat{}\{2+\}$ reabsorption
- PTH stimulates kidney production of active vitamin D (1,25-dihydroxyvitamin D_3)
- These actions raise blood $\mathrm{Ca}\hat{}\{2+\}$ back to normal
- Normal $\mathrm{Ca}\hat{}\{2+\}$ inhibits further PTH secretion (negative feedback)

Osteoporosis

- **Osteoporosis** (骨质疏松症): Progressive loss of bone mass
- Bone resorption exceeds bone deposition
- Results in fragile bones with increased fracture risk
- Common in elderly, especially postmenopausal women
- Prevention: Adequate calcium/vitamin D, exercise, avoid smoking/excessive alcohol

Osteoporosis

- Scanning electron microscopy reveals dramatic changes in bone microarchitecture
- **Normal bone**: Thick, dense trabeculae (骨小梁) with good connectivity
- **Osteoporotic bone**: Thin, fragile trabeculae with many gaps and poor connectivity
- This loss of structural integrity greatly increases fracture risk

Osteoporosis

- Treatment includes calcium, vitamin D, bisphosphonates, and other medications

Vitamin D Production and Activation

- **Vitamin D** (维生素 D) is actually a prehormone, not a true vitamin
- Produced in skin from cholesterol upon UV exposure, or obtained from diet
- Requires two hydroxylation steps to become active:
  1. Liver: Adds -OH at position 25 → 25-hydroxyvitamin D_3
  2. Kidney: Adds -OH at position 1 → 1,25-dihydroxyvitamin D_3 (active form)
- Final step is stimulated by PTH and low phosphate

Vitamin D Production and Activation

- Active vitamin D production involves sequential conversions in multiple organs:
- Skin/diet provides vitamin D_3 (cholecalciferol, 胆钙化醇)
- Liver hydroxylates to form 25-hydroxyvitamin D_3
- Kidneys perform final hydroxylation to form 1,25-dihydroxyvitamin D_3 (**calcitriol**, 骨化三醇)
- This final step is regulated by PTH, making vitamin D part of calcium homeostasis system

Vitamin D Production and Activation

- Active vitamin D then increases intestinal calcium absorption

Integrated Calcium Homeostasis

- Multiple hormones coordinate to maintain calcium balance
- PTH and vitamin D work together synergistically
- System maintains $\mathrm{Ca}\hat{\ }\{2+\}$ while preventing $PO_4\hat{\ }\{3\text{-}\}$ accumulation
- Kidneys play key role by selectively excreting phosphate

Integrated Calcium Homeostasis

- Integrated negative feedback system maintains calcium homeostasis:
- When Ca^{2+} falls: PTH ↑, stimulates bone resorption and kidney reabsorption
- PTH stimulates 1,25-dihydroxyvitamin D_3 production in kidneys
- Vitamin D increases intestinal Ca^{2+} and PO_4^{3-} absorption
- Kidney excretes excess PO_4^{3-} while retaining Ca^{2+}
- Net result: Blood Ca^{2+} rises to normal, PO_4^{3-} stays balanced

Calcitonin: Antagonist to PTH

- **Calcitonin** (降钙素): Secreted by parafollicular cells (C cells, 滤泡旁细胞) of thyroid
- Secretion stimulated by high blood $\mathrm{Ca}\hat{}\{2+\}$
- Actions opposite to PTH:
  - Inhibits osteoclast activity (reduces bone resorption)
  - Increases urinary excretion of $\mathrm{Ca}\hat{}\{2+\}$ and $\mathrm{PO_4}\hat{}\{3\text{-}\}$
- Physiological importance in humans is unclear

Calcitonin: Antagonist to PTH

- Calcitonin provides a negative feedback response to high calcium:
- Elevated blood Ca^{2+} stimulates calcitonin secretion
- Calcitonin inhibits osteoclast activity, reducing bone resorption
- Calcitonin increases renal excretion of Ca^{2+} and PO_4^{3-}
- Net effect: Lowers blood calcium levels
- Action is antagonistic to PTH, but its physiological role in adults is less clear

Summary of Hormonal Control

- Calcium and phosphate balance involves coordinated hormonal regulation
- Three major hormones: PTH, vitamin D, and calcitonin
- Act on three target tissues: intestine, kidney, and bone
- System maintains tight control of blood calcium despite variable intake and losses

Summary of Hormonal Control

- Comprehensive summary of hormonal effects on calcium and phosphate homeostasis
- **PTH**: No direct intestinal effect, increases kidney $\mathrm{Ca}\hat{}\{2+\}$ reabsorption, stimulates bone resorption
- **1,25-dihydroxyvitamin D_3**: Increases intestinal absorption, increases kidney reabsorption, stimulates bone resorption

Summary of Hormonal Control

- **Calcitonin**: No intestinal effect, decreases kidney reabsorption, inhibits bone resorption
- Understanding these interactions is essential for treating calcium disorders